

## Section 4.2(b)- Expected value and Variance for continuous random variable

# Expected value and variance of a continuous random variable

Given continuous random variable  $T$  with probability density function  $f(t)$  we have

$$\mu = E[T] = \int_{-\infty}^{\infty} t \cdot f(t) dt$$

$$E[g(T)] = \int_{-\infty}^{\infty} g(t) \cdot f(t) dt$$

$$\sigma^2 = \text{Var}[T] = \int_{-\infty}^{\infty} (t - \mu)^2 \cdot f(t) dt \text{ or}$$

$$\sigma^2 = E[T^2] - \mu^2$$

$$SD(T) = \sqrt{\text{Var}(X)}$$

# Review: Properties of expected value and variance

Everything in this slide is true for both continuous and discrete random variables.

Properties of Expectation: Expectation is always a linear function meaning:

(a)  $E[X + Y] = E[X] + E[Y]$ , for any random variables  $X$  and  $Y$ .

(b)  $E[aX] = a \cdot E[X]$

(c)  $Var(aX) = a^2 Var(X)$

**Eg1.** For a particular machine, its useful lifetime (random variable  $T$ ) is modeled by

$$f(t) = \begin{cases} 2e^{-2t} & t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Find  $E[T]$  and  $\text{Var}(T)$ .

$$E[T] = \int_{-\infty}^{\infty} t f(t) dt = \int_0^{\infty} 2t e^{-2t} dt$$

using integration by parts

$$E[T] = -t e^{-2t} - \frac{e^{-2t}}{2} \Big|_0^{\infty}$$

|               | $\frac{d}{dt}$ | $\int$              |
|---------------|----------------|---------------------|
| $\oplus$ $t$  | $\searrow$     | $2e^{-2t}$          |
| $\ominus$ $1$ | $\searrow$     | $-e^{-2t}$          |
| $\oplus$ $0$  | $\searrow$     | $\frac{e^{-2t}}{2}$ |

$$E[T] = -\lim_{t \rightarrow \infty} t e^{-2t} + \int_0^{\infty} -\frac{e^{-2t}}{2} + \frac{e^0}{2}$$

$$E[T] = -\lim_{t \rightarrow \infty} \frac{t}{e^{2t}} + \frac{1}{2} \stackrel{\text{L'Hopital}}{=} -\lim_{t \rightarrow \infty} \frac{1}{2e^{2t}} + \frac{1}{2} = \frac{1}{\infty} + \frac{1}{2} = \frac{1}{2}$$

$$\text{Var}(T) = E[T^2] - (E[T])^2$$

$$E[T^2] = \int_0^{\infty} t^2 f(t) dt = \int_0^{\infty} 2t^2 e^{-2t} dt$$

$$E[T^2] = -t^2 e^{-2t} - 2t \frac{e^{-2t}}{2} - \frac{e^{-2t}}{2} \Big|_0^{\infty}$$

$$E[T^2] = -e^{-2t} \left[ t^2 + t + \frac{1}{2} \right] \Big|_0^{\infty} = -\frac{t^2 + t + \frac{1}{2}}{e^{2t}} \Big|_0^{\infty} = -\lim_{t \rightarrow \infty} \frac{t^2 + t + \frac{1}{2}}{e^{2t}} + \frac{1}{2} e^0$$

|   | 0     | ∫                     |
|---|-------|-----------------------|
| + | $t^2$ | $\frac{1}{2} e^{-2t}$ |
| - | $2t$  | $-e^{-2t}$            |
| + | $2$   | $\frac{e^{-2t}}{2}$   |
| - | $0$   | $-\frac{e^{-2t}}{4}$  |

$$\begin{aligned}
 E[T^2] &= \lim_{t \rightarrow \infty} \frac{2t+1}{2e^{2t}} + \frac{1}{2} \stackrel{\text{L'Hopital}}{=} \lim_{t \rightarrow \infty} \frac{2}{4e^{2t}} + \frac{1}{2} = -\frac{1}{\infty} + \frac{1}{2} \\
 &= 0 + \frac{1}{2} \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\text{Var}(T) = E[T^2] - (E[T])^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

**Note:** We will learn that later that a random variable  $T$  with pdf  $f(t) = \lambda e^{-\lambda t}; t \geq 0$  is exponentially distributed with parameter  $\lambda$  and

$$\text{Var}(T) = \frac{1}{\lambda^2} \text{ and } E[T^2] = \frac{1}{\lambda^2}$$

Eg 2. Given the probability density function

$$f(x) = \begin{cases} 6x(1-x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find  $E[X]$  and  $\text{Var}(X)$ .

$$\mu = E[X] = \int_0^1 x f(x) dx = \int_0^1 6x^2(1-x) dx$$

$$\mu = \int_0^1 6x^2 - 6x^3 dx$$

$$\mu = 2x^3 - \frac{6x^4}{4} \Big|_0^1 = 2 - \frac{3}{2} = \frac{1}{2}$$

$$E[X^2] = \int_0^1 x^2 f(x) dx = \int_0^1 6x^3(1-x) dx = \int_0^1 (6x^3 - 6x^4) dx$$

$$= \left. \frac{6x^4}{4} - \frac{6x^5}{5} \right|_0^1 = \frac{6}{4} - \frac{6}{5} = 6 \cdot \frac{1}{20} = \frac{3}{10}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{3}{10} - \frac{1}{4} = \frac{6-5}{20} = \frac{1}{20}$$